

Vivek Chavan | PhD Candidate at TU Berlin and Researcher at Fraunhofer IPK

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SUMMARY: Machine Learning, Computer Vision and Robotics researcher developing multimodal learning systems for long-horizon procedural understanding and embodied action. Research on *Multimodal Ego/Exo Perception, Vision-Language(-Action) Models, Robot Learning, and Continual Learning*.

Expected graduation: March 2028 | Available full-time for research internships throughout 2027.

WORK EXPERIENCE

Research Associate | Fraunhofer IPK, Berlin

Nov 2022 - present

Lead research and development in Computer Vision, multimodal learning, and robotics, spanning data-centric learning, representation learning, ego/exo perception, long-horizon procedural understanding, and embodied AI. Developed and deployed ML systems for classification, segmentation, quality assessment, multimodal perception, and MLOps; supervised students and research interns; led international research collaborations and coordinated €3M+ in research funding proposals (€1.8M awarded as of Q2 2026).

Intern and Student Assistant | Fraunhofer IPK, Berlin

Oct 2021 - Oct 2022

Developed and evaluated Computer Vision and deep learning methods for industrial applications. Curated two research datasets and investigated data quality bottlenecks affecting model robustness and generalization.

CAE Engineer | Tata Technologies & Jaguar Land Rover, India & UK

July 2014 – Aug 2019

Developed large-scale CFD/CAE simulations for vehicle performance; co-authored two technical papers introducing new modelling methods; collaborated with international engineering teams for system-level optimization.

EDUCATION

2023 - Present PhD at *Technical University of Berlin*.

Continual Ego/Exo Machine Perception: From Procedural Understanding to Embodied Action
[Nominated for the *Google PhD Fellowship* and the *Amazon Research Grant*]

2019 - 2022 Master's Degree at *RWTH Aachen University*.

Thesis: Green Incremental Learning.

[Thesis Grade: 1,0 (Perfect Score)]

2010 - 2014 Bachelor's Degree at *Pune University, India*.

[Graduated Top of the Class]

SELECTED FIRST AUTHOR PUBLICATIONS

[GOOGLE SCHOLAR](#)

V. Chavan, Y. Imgrund, T. Dao, et al., “IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants.” *Advances in Neural Information Processing Systems (NeurIPS 2025)*.

[Established benchmarks and baselines for Mistake Detection, Video Question Answering, collaborative work and task summarization in human-centric industrial settings.]

V. Chavan, J. Krüger, “AI-based worker guidance in assembly and disassembly operations using multimodal ego/exo-centric data capture and structured task knowledge.” *CIRP Annals - Manufacturing Technology, 2026*.

[Encoded temporal multimodal information from egocentric expert demonstrations into structured representations. Enabled context-aware worker guidance and procedural documentation.]

V. Chavan, P. Koch, M. Schlüter, et al., “Towards Realistic Evaluation of Industrial Continual Learning Scenarios with an Emphasis on Energy Consumption and Computational Footprint.” *Proc. IEEE/CVF International Conference on Computer Vision (ICCV 2023)*.

[Established evaluation protocols for incremental learning by integrating energy consumption and computational load as primary metrics for real-world adoption.]

V. Chavan, P. Koch, M. Schlüter, et al., “Active Data Collection and Management for Real-World Continual Learning via Pretrained Oracle.” *Proc. IEEE/CVF Computer Vision and Pattern Recognition Workshops (CVPRW 2024)*.

[Developed a framework for active coreset selection and distribution matching in non-curated streams using a pretrained oracle. Demonstrated improvements over state-of-the-art exemplar selection strategies.]

V. Chavan, O. Heimann, J. Krüger, “On the Application of Egocentric Computer Vision to Industrial Inspection.” *Proc. European Conference on Computer Vision Workshops (ECCVW 2024)*.

[Fused narration, gaze, and hand-pose for automated annotation in industrial inspection via a multimodal pipeline. Benchmarked egocentric perception and quantified ego-to-exocentric domain shifts.]

V. Chavan, O. Heimann, J. Krüger, “A System 1 and System 2 Perspective on Continual Learning for Practical Implementation.” *Proc. European Conference on Computer Vision Workshops (ECCVW 2024)*.

[Integrated contextual semantic encoding with robust pretraining to enhance model generalization and mitigate catastrophic forgetting across varied data distributions.]

V. Chavan, P. Xie, Y. Shi, O. Heimann, K. Haninger, J. Krüger, “What Matters, When? Improving Object Selectivity and Conditional Visual Grounding in Visuomotor Imitation Policies.” *Under review at IEEE Robotics and Automation Letters (RA-L)*.

[We diagnose how imitation policies can lose task-relevance under unseen conditions and propose a generalizable corrective framework that yields more robust, data-efficient behavior across ACT and pretrained VLA policies.]

V. Chavan, Y. Shi, K. Haninger, O. Heimann, J. Krüger, “Towards Neuro-Symbolic Procedural Reasoning for Long-Horizon Vision-Language-Action Manipulation.” *Oral presentation at the ECCV 2026 X-Reason Workshop; extended full paper in preparation for CVPR 2027*.

[We augment VLA policies with structured task reasoning and multimodal procedural memory, enabling them to maintain task state, resolve action dependencies, and execute visually grounded long-horizon manipulation.]

OPEN-SOURCE RESEARCH DATASETS

STEP-ERR [2026]: Benchmark of paired correct and erroneous long-horizon robot executions with step-level task-state annotations for error detection, temporal localization, procedural reasoning, state tracking, and recovery. (Initial presentation at ECCV Workshop 2026; extended benchmark targeting CVPR 2027.)

IndEgo [2025]: A large-scale multimodal ego/exo dataset capturing RGB, time-synchronized gaze, motion data in procedural tasks and diverse industrial use-cases (e.g., Assembly/disassembly, logistics, repair).

MVIP [2024]: Multimodal RGB-D benchmark of 300+ industrial parts across 10 calibrated views. Includes semantic masks, physical properties, and text metadata; utilized for modality fusion and few-shot transfer.

InVar-100 [2023]: Industrial object recognition benchmark containing 100 distinct object classes across varied environmental contexts. Simulates the distribution shift and visual noise characteristic of industrial deployments.

ACADEMIC SERVICE AND LEADERSHIP

Reviewer and Sub-reviewer: CVPR 2026, ECCV 2026, and NeurIPS 2026 (invited)

Mentorship: Supervised 5 Bachelor/Master theses, 9 research internships, and 9 student semester project participants at Fraunhofer IPK/TU Berlin.

AI Grid Representative (Berlin Area): Coordinating interdisciplinary research, networking, and knowledge exchange for the German national AI initiative. Co-authored two technical reports on *the broader impact of AI*.

CERTIFICATIONS

NVIDIA Multimodal AI Agents & RAG Pipelines for Production; Coursera MLOps Specialization; TensorFlow Developer Certificate; Coursera Deep Learning & TensorFlow Specializations; Lean Six Sigma Yellow Belt.

SKILLS & INTERESTS

Programming & Tools: Python, PyTorch, JAX, TensorFlow, ROS/ROS 2, Linux, Git, MLOps.

Research: Computer Vision, Multimodal Learning, Vision-Language(-Action) Models, Robot Learning, Egocentric/Exocentric Vision, Continual Learning, Data-Centric Learning.

Interests: Scientific Writing, Public Speaking, Fitness and Wellbeing.

Vivek Chavan